Architecture and Requirements for Observability, C&I of Network Management Agents

An architecture and requirements briefing for supervising autonomous AI agents deployed for network operations

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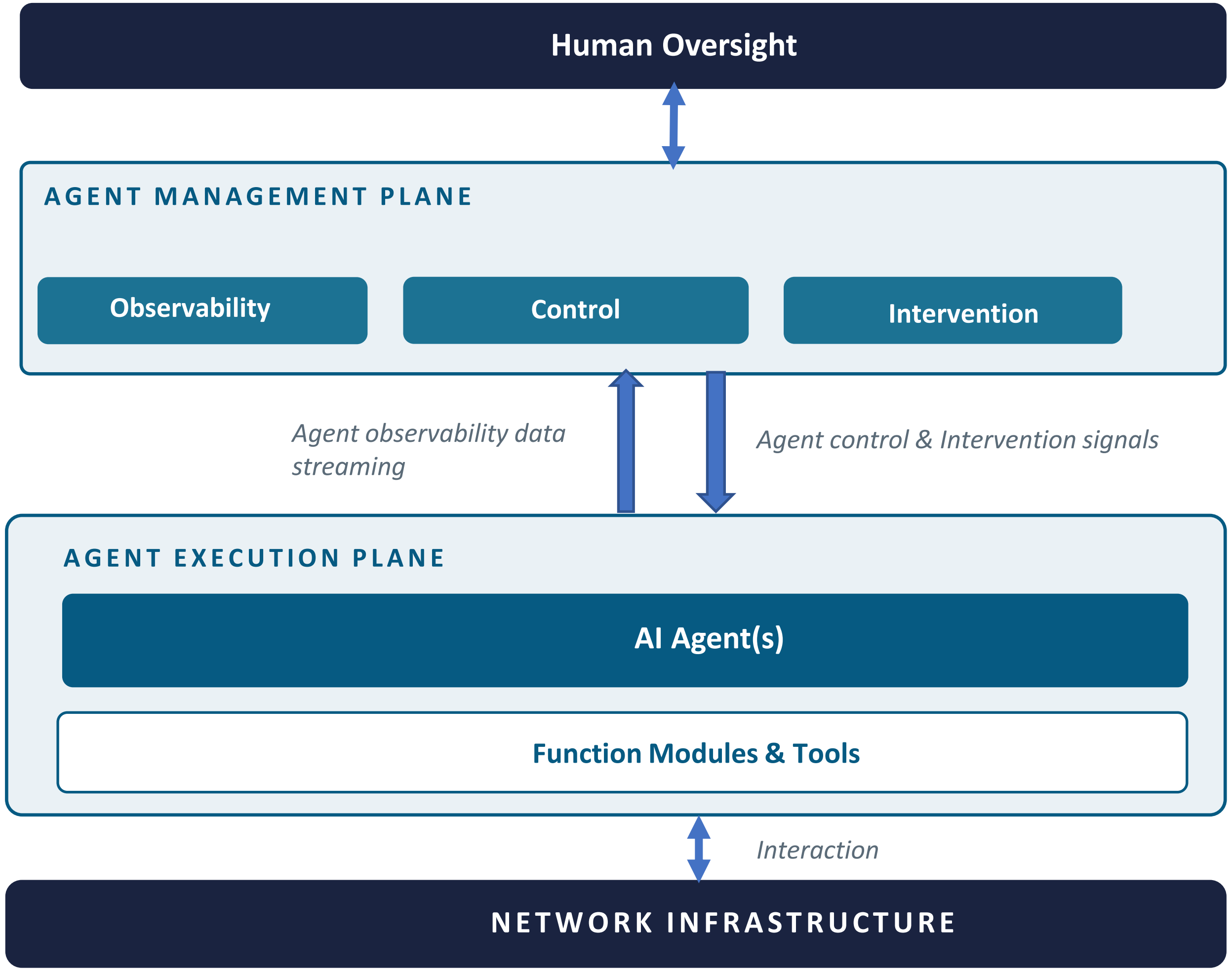
https://datatracker.ietf.org/doc/draft-mcw-opsawg-icon-requirements/

Scope & Positioning

- Architecture Design
  - A high-level interaction framework that outlines the boundaries and scope.
- Requirements between the agent and the entity performing the agent observability, control, and intervention
  - Motivated by draft-wnd-opsawg-icon-ps which details the operational gaps
  - Might not be fully comprehensive, but hopefully establish a set of essentials that can be expanded in the future
- Solution-neutral
  - Protocols, data models, or API choices are intentionally left for future work.

A Reference Architecture



Human Oversight - the top-level authority

- Agent runtime monitoring
- Policy & constraint injection
- Emergency intervention trigger
- Escalation request handling

Agent Management Plane – centralized layer for agent assurance

- Agent runtime telemetry data collection
- Control policy validation and distribution
- Agent runtime intervention instruction (e.g., kill switch, pause/resume, or rollback)

Agent Execution Plane – agent runtime environment

- Network management tasks receiving
- LLM reasoning
- Policy-compliant network operation task execution
- Execution result reporting

Deriving Requirements from Problem Statements-Agent Observability

See what agents did, why they did it, and how they're performing.

Problem Statement Gap I-D. wnd-opsawg-icon-ps	Essential Requirement I-D. mcw-opsawg-icon-requirements	Requirement Description
3.1.1 Limited Transparency in Planning and Decision-Making	OBS-1: Execution Trajectory Capture	The full path visibility, including the plan, the invoked tools/skills, generated configuration change, and the resulting operational state
5.1 OpenTelemetry only captures the execution process, not the operational motivation.	OBS-2: Reasoning Provenance Capture	Exposes why a specific decision was made — the network operational signals (e.g., alarms, incidents, telemetry streams) and confidence level behind a specific network action.
5.1 lacks native support for observing metrics such as an agent's reasoning logic and internal confidence levels	OBS-3: Operational Metrics Collection	Tracks an agent's operational health and performance when performing specific network operations.
3.1.2. Ambiguity of Accountability Attribution	OBS-4: Auditability and Accountability	Immutable audit logging to allows the attribution of failures to responsible entities (e.g., intent interpretation, LLM reasoning, tool invocation)

Deriving Requirements from Problem Statements- Agent Control

Steer agent behaviors as much as possible within the anticipated scope.

Problem Statement Gap I-D. wnd-opsawg-icon-ps	Essential Requirement I-D. mcw-opsawg-icon-requirements	Description
3.2.1. Inadequacy of Deterministic Constraints	CTL-1: Intent Validation & Alignment	Requires the agent to check that what it is about to act actually matches the operational intent it was given (e.g., mandate the validation from a digital twin simulation platform before config changes).
3.2.1. Inadequacy of Deterministic Constraints	CTL-2: Temporal Validity	Constraints an agent's actions to a valid operating window(e.g., only allows the config change from 10PM to 11PM).
3.2.2 Static IAM Limitation	CTL-3: Access & Permission	Granular access control to: • specific network domains/areas, devices; • APIs and tools; • Data and operations on a specific NE (e.g., datastores, modules, nodes, RPCs.)
3.2.2 Static IAM Limitation	CTL-4: Authorization & Approval	Specification of some network operations as requiring a human decision before execution.
3.2.2. Static IAM Limitation	CTL-5: Dynamic Boundary Adaptation	Enable dynamic adjustment of operational bounds (e.g., tighten the agent's permissible access from RW to RO) based on the network's current operational state.

Deriving Requirements from Problem Statements-Agent Intervention

Emergency runtime interruption when the agent deviates from expected behaviors.

Problem Statement Gap I-D. wnd-opsawg-icon-ps	Essential Requirement I-D. mcw-opsawg-icon-requirements	Description
3.3.1 lack of human oversight Essential human oversight capabilities including execution interruption 5.5 absence of an interface that allows an authorized authority outside the framework or outside the agent application to signal intervention; current practice of intervention lacks soft redirect, scope restriction, checkpoint, task suspension, rollback, hard termination	INT-1: Execution Interruption	stop or redirect an agent's execution. Guarantees an agent can be paused/terminated from outside its own decision loop, even if the agent itself is unresponsive or stuck.
3.3.1. Lack of Human Oversight Essential human oversight capabilities including deterministic transaction rollback and Recovery	INT-2: Rollback and Recovery	Able to reverse the operations performed by the agent on the network based on different rollback granularities: workflow level, task level, and context level.
3.3.1. Lack of Human Oversight Essential human oversight capabilities including human escalation	INT-3: Escalation	Moves a decision the current agent can't or shouldn't make alone up to someone with broader authority, without losing operational context.
4.6 Intervention and Control is concerned with ensuring actions can be corrected or reversed when necessary.	INT-4: Correction	Allows a supervisor to fix what went wrong, e.g., the intent, the data, or the plan instead of only being able to accept or reject the agent's output.

Comments, Questions, Concerns?